

Comprehensive Examination

First Semester 2023-2024

Date: 9th Dec 2023

Time: 180min

CLOSED BOOK

Max Marks: 80

Name:

ID:

Recheck Request:

Solutions

Instructions:

- Please answer only in the space provided. Anything written outside will not be evaluated.
- Clearly show all steps used to arrive at final answer. Final answer needs to be boxed.
- Anything written in pencil will not be evaluated and will not be accepted for recheck
- Partial marking is at the discretion of the instructor.
- Please note that answers without proper units will be marked wrong.
- Overwriting will not be evaluated. Please keep your work legible, clean and crisp.

Marks obtained:

Question No.	Marks
1	
2	
3	
4	
5	
6	
7	
8	
Total	

Question 1 (9 marks)

A 15X inverter is driving a 3X inverter at the end of a long wire in a $0.5\mu m$ ($L = 0.5\mu m$) technology. The Gate capacitance and the diffusion capacitance C per unit width, in this technology is $2fF/\mu m$. The effective resistance per unit width of the NMOS R , is $2.5 k\Omega \cdot \mu m$. The smallest inverter is sized with $W_p/W_n = 2/1$ for equal rise and fall delays. Assuming that the wire resistance is $0.1 \Omega\mu m^{-1}$ and the wire capacitance is $0.2 fF/\mu m$, find the length of the wire that can be connected between the two so that the propagation delay does not exceed 100ps.

For the 1x inverter $R = 5 k\Omega$
 $C = 3 fF$
 $l \leq 173.15 \mu m$

Question 2 (12 marks)

Consider a CMOS inverter in a $0.8\mu m$ technology ($L = 0.8\mu m$) with the following parameters ($\lambda = 0$).

nMOS	$V_{Tn} = 0.5 V$	$\mu_n C_{ox} = 98 \mu A/V^2$
pMOS	$V_{Tp} = -0.48 V$	$\mu_p C_{ox} = 46 \mu A/V^2$

The inverter is designed for a switching threshold of 0.6 V at a $V_{DD} = 1.2 V$. The output capacitance of such an inverter is given by $C_{out} = 5fF + C_{db,n} + C_{db,p}$. The drain capacitances of the NMOS and the PMOS are given by $C_{db,n} = 0.16fF + 1.7W_n$ and $C_{db,p} = 0.13fF + 1.4W_p$ where W_n and W_p are in μm . Determine the channel widths so that $\tau_{pHL}(50\%)$ is less than 90ps.

$$W_n = 7.29 \mu m$$

$$W_p = 10.79 \mu m$$

Question 3 (5+4 marks)

Design a CMOS static gate to compute $F = (A + B) \cdot (C + D)$ with least delay. (No additional Boolean simplification needed). Each input can present a maximum input capacitance of $10C_{inv}$. The output must drive a load of $160 * C_{inv}$. C_{inv} denotes the input capacitance of the smallest 2/1 inverter. Choose transistor sizes to achieve least delay.

2 stage design.

Gate 1 $(A+B)(C+D)$ + $\rightarrow$

20P	56.54P
10N	28.27N

Design a domino gate to achieve the same function. You can skew the gates so that non-critical transition has 2X the resistance of the critical one.

unfooted design

$(A+B)(C+D)$ + $\rightarrow$

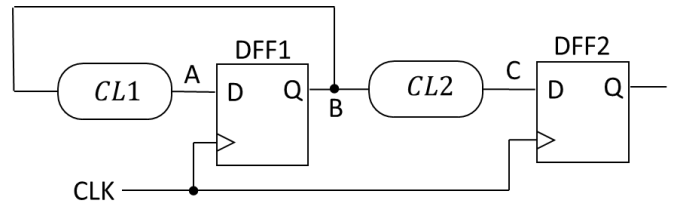
NMOS = 30	10 skew
	N = 90
	P = 90

Question 4 (3+3+3 marks)

The following FSM has been designed with an initial clock period is 9ns. The D-FF have the following parameters: $t_{ccq} = 2ns$, $t_{pcq} = 3ns$, $t_{setup} = 1ns$ and $t_{hold} = 5ns$. The combinatorial blocks have the following timing parameters: CL1: $t_{cd} = 2ns$, $t_{pd} = 3ns$; CL2: $t_{cd} = 4ns$, $t_{pd} = 5ns$. A, B and C are node names to track the signal.

Is there a timing violation in the circuit? If so what is it and on what path does it occur?

B-CL1-A - Hold-time violation



Assume that you have a non-inverting buffer with $t_{cd} = 2ns$, $t_{pd} = 4ns$. Draw the new circuit diagram showing how these may be used to fix the timing violation.

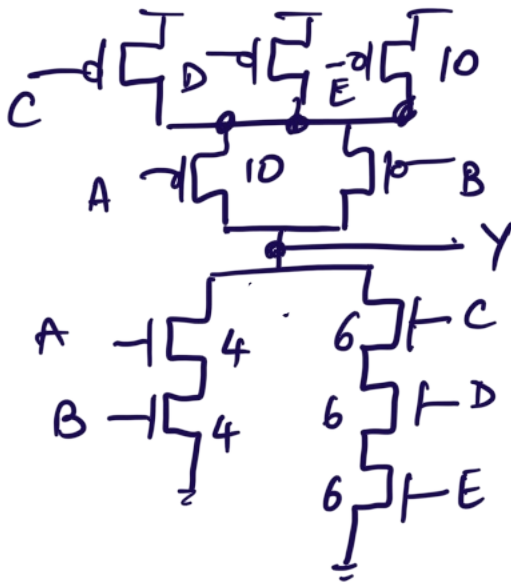
Add 1 buffer in the path

What is the final clock period after the fix is put in place.

$$T_{clk} = 12ns$$

Question 5 (5+6 marks)

Design a static CMOS gate to implement the function $F = \overline{A \cdot B + C \cdot D \cdot E}$. Size the transistors so that the worst-case pull-up and pull-down resistance of the gate is equal to that of the smallest inverter in the technology with P/N sizes of 2/1. If there are multiple ways of designing, pick a design with the least nominal parasitic delay.



In this technology
 $R_p = 2.5 R_n$
 Since it is a 5/2 inverter

Calculate the Elmore delay for the following transitions. The input word is given by ABCDE. You may assume that no source/drain sharing is allowed.

(00101) → (00111)

$44RC$ $R = NMOS R$

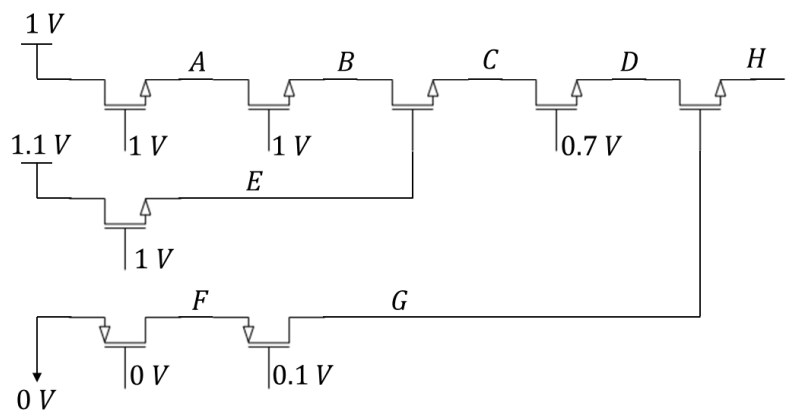
(00111) → (00011)

$35RC$

Question 6 (8 marks)

Find the voltages at the various nodes in the circuit. Assume $V_{Tn} = 0.2 \text{ V}$ and $V_{Tp} = -0.3 \text{ V}$

A	0.8 V
B	0.8 V
C	0.6 V
D	0.5 V
E	0.8 V
F	0.3 V
G	0.4 V
H	0.2 V



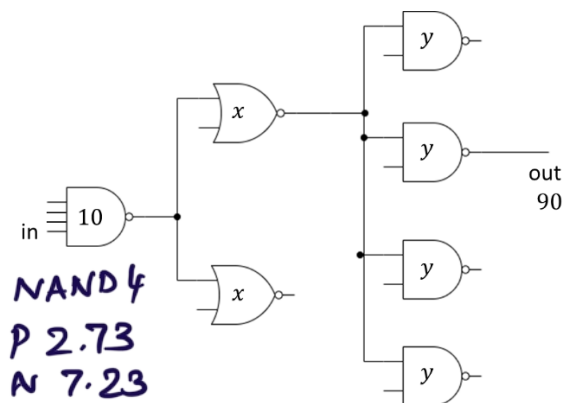
Question 7 (12 marks)

Consider the following circuit. The input NAND has an input capacitance of 10 units. The output capacitance is 90 units where 3.5 units denotes the gate capacitance of the smallest inverter in the technology sized 1.5/1. Assume that the polarity of the output is not important.

Calculate the path effort in the critical path of the circuit.

$$F = 354.82$$

$$\hat{f} = 7.079 \quad D = 29.237$$



Size the critical path for minimum delay. Show the transistor level sizing of the gates.

$$y = 17.8 < 7.64 \text{ P}$$

$$a = 16 < 10.16 \text{ N}$$

$$area = 246.4$$

Can this design be improved for smaller delay? If so suggest the modification and resize the path

4 stages — good trade off of delay and area

$$f = 4.34$$

$$D = 26.36$$

$$Z_{inv} = 20.73$$

$$y = 6.69$$

$$x = 7.04$$

What is the % improvement ($\frac{old-new}{old}$) in delay you were able to achieve?

$$9.84\%$$

Approximating the circuit area to the total gate area what is the estimated % increase in the circuit area?

$$\left(\frac{new-old}{old}\right)$$

$$6.5\%$$

Question 8 (2+1+2+2+1+2 marks)

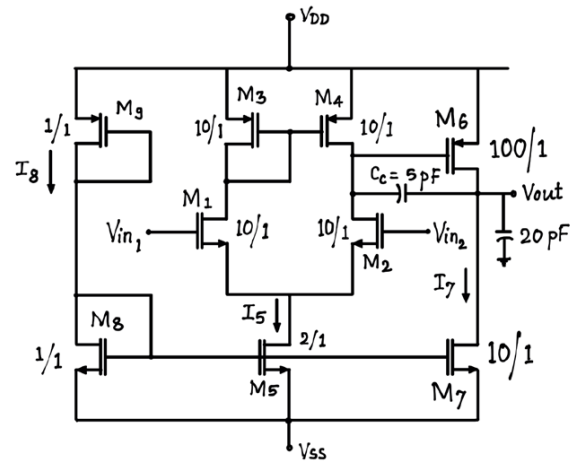
For the operational amplifier given below, assume all transistors are operating in saturation region. The open loop unity gain bandwidth is 1 MHz. The W/L ratios given in the figure are in $\mu m/\mu m$. The MOSFET parameters are as follows: $V_{Tn} = |V_{Tp}| = 0.7 V$, $\mu_n C_{ox} = 110 \mu A/V^2$, $\mu_p C_{ox} = 50 \mu A/V^2$, $\lambda_n = 0.04 V^{-1}$ and $\lambda_p = 0.05 V^{-1}$ for $L = 1 \mu m$, $V_{DD} = 2.5 V$ and $V_{SS} = -2.5 V$

Find the DC value of I_5 , I_7 and I_8 .

$$I_8 = 115.60 \mu A$$

$$I_5 = 231.2 \mu A$$

$$I_7 = 1.156 mA$$



The low frequency (DC) differential voltage gain

$$1581.06 V/V$$

The Gain-Bandwidth Product (GBP) in Hz

$$16.04 MHz$$

Slew rate

$$46.26 \text{ V}/\mu\text{s}$$

Power Dissipation

$$7.514 \text{ mW}$$

Phase Margin

$$50.91^\circ$$